

Joshua Plummer | Curriculum Vitae

Montgomery, Alabama | plummerjrandall@gmail.com

Professional Profile

Teaching-focused mathematics educator and curriculum developer expanding expertise into instructional technology, professional development, and modern learning systems. Grounded in adult learning theory and evidence-based course design, with a strong record of creating engaging, student-centered learning experiences. Recognized through repeated student-nominated teaching honors.

Education

Auburn University at Montgomery

Ed.D. Instructional Systems and Learning Sciences
Concentration: Instructional Technology

Exp. May 2028

University of South Alabama

M.S. Mathematics

Aug. 2016 – May 2018

B.S. Mathematics & Statistics

Aug. 2010 – May 2016

B.S. Physics

Aug. 2010 – May 2016

Experience

Auburn University at Montgomery

Lecturer

Aug. 2024 – Present

- Design and teach engaging, equitable, and outcomes-aligned curricula for gateway mathematics and statistics courses by utilizing collaborative and experiential approaches.
- Collaborate with committee peers to coordinate syllabi, assessments, and corequisite support for multi-section gateway courses.
- Review course-performance data, including student evaluations, DFW patterns, and informal surveys to refine instruction and student support.

St. Luke's Episcopal School

Upper School Teacher

Aug. 2019 – Apr. 2023

- Taught advanced and AP-level mathematics and science courses.
- Established AP Calculus AB into the upper-school curriculum.
- Applied the Experience First, Formalize Later model to promote mathematical intuition.
- Spearheaded the development of inquiry-based, active-learning curricula across departments.
- Drove departmental adoption of standards-based grading and active-learning practices.
- Sponsored the Mu Alpha Theta chapter.
- Coached the school esports team.

Faulkner University*Adjunct Instructor**Jan. 2021 – May 2022*

- Taught undergraduate mathematics courses in face-to-face format.
- Develop department curricula and adopt textbooks.
- Assessed student work and delivered timely feedback to support skill development and course success.

University of South Alabama*Adjunct Instructor**Aug. 2018 – Dec. 2022*

- Taught undergraduate mathematics courses in face-to-face format.
- Developed curricula to engage students and maximize learning.
- Held office hours and one-on-one tutoring sessions.

University of South Alabama*Graduate Teaching Assistant**Aug. 2016 – May 2018*

- Led recitation and problem-solving sessions.
- Developed and graded weekly quizzes.
- Held weekly office hours and exam review sessions.

University of South Alabama*Lead Peer Academic Leader (PAL)**Aug. 2012 – May 2016*

- Mentored students and connected them with academic-support resources.
- Supervised and evaluated a team of 5–6 PALs.
- Supported onboarding, performance reviews, interviewing, and selection of PALs.

Instructional Design, Curriculum, and Faculty Development

Selected work in curriculum design, faculty development, and instructional systems.

Professional Learning Toolkit*Auburn University at Montgomery**2026**Accessible & Rigorous: Designing Mathematics Instruction for the Modern Learner*

- Designed a professional-learning toolkit for mathematics faculty, grounded in adult learning theory, backward design, and accessibility with three job-embedded professional development sessions.
- Facilitated a one-on-one adaptation of the *Your Page, Rebuilt* professional development session, guiding a colleague through an accessible Canvas page redesign and used feedback to refine the session.

Co-requisite Workbook Development*Auburn University at Montgomery**2025*

- Revised and redeveloped co-requisite workbooks for Elementary Statistics, Precalculus, and College Algebra.
- Reviewed content, refined explanations, and restructured activities for clarity and alignment with learning objectives.

Academic Materials Development

University of South Alabama

2015

- Spearheaded the project with the Office of Student Success to support students in historically challenging courses in mathematics, physics, and chemistry.
- Developed 100+ academic-support resources, including quick-reference guides and example videos.

Leadership and Service

2026 – Present: Social Committee	<i>Chair</i>
2024 – Present: Course Coordination Committees	<i>Member</i>
2024 – Present: Co-requisite Support Committees	<i>Member</i>
2026 – Present: Online Course Integrity Committee	<i>Member</i>
2026 – Present: Hiring Committee	<i>Member</i>
2025 – Present: Alabama State-Wide High School Mathematics Contest	<i>Member</i>
2025 – Present: SK-Day Committee	<i>Member</i>
2025 – 2026: Social Committee	<i>Co-Chair</i>

Certifications and Professional Development

Google Data Analytics

Coursera

in progress

Google AI Fundamentals

Coursera

2026

Elements of AI: Introduction to AI and Building AI

ElementsofAI.com

2026

AI for Educators

Microsoft

2026

Generative AI for Educators with Gemini

Google

2026

NextGen Learning: AI Integration Series

Purdue University Global

2026

Certificate Series in Accessibility Practices for Microsoft Suite and Canvas

Auburn University at Montgomery

2026

Enhancing Online Education through Instructor-Created Video Content

Purdue University Global

2026

Flight School (Student Success & Peer Mentoring Training)

Auburn University at Montgomery

2026

Certificate in Applied Data Analytics

Auburn University at Montgomery

2025

HHMI Grant-Supported Teaching & Student Success Training <i>Auburn University at Montgomery</i>	2024
Experience First, Formalize Later Workshop <i>Math Medic Foundation</i>	2022
AP Summer Institute: AP Calculus AB <i>Auburn University</i>	2020

Selected Awards and Honors

Student-Nominated Significant Impact Certificate of Appreciation <i>Auburn University at Montgomery</i>	2026
<i>Auburn University at Montgomery</i>	2025
<i>Auburn University at Montgomery</i>	2024
Conference of Southern Graduate Schools Outstanding Thesis Award <i>University of South Alabama</i>	2019
Sandra & Melton McLaurin Graduate Fellowship <i>University of South Alabama</i>	2017
Recognition of Dedication and Leadership as a Lead JagPAL <i>University of South Alabama</i>	2016
Chris Nash Mathematics Competition Scholarship <i>University of South Alabama</i>	2015
Induction into Sigma Pi Sigma <i>University of South Alabama</i>	2013

Research and Publications

Graduate Thesis <i>University of South Alabama</i>	2017 – 2018
<i>On Matroid Representability</i> (M.S. thesis, advisor: Dr. Elena Pavelescu). Examined matroid representations, including Rota's conjecture, with original proofs of selected results.	
Editorial on the Importance of General Education Requirements <i>University of South Alabama</i>	2016
Plummer, J. (2016). <i>The Importance of General Education</i> . <i>The Vanguard</i> , 59(9), 7. Written for a freshman seminar course.	
Astronomy Research and Publication <i>University of South Alabama</i>	2014 – 2015
Studied visual binary stars via speckle interferometry under Dr. Kent Clark, part of a study led by Russell Genet. Genet, R., et al. (2015). <i>Kitt Peak Speckle Interferometry of Close Visual Binary Stars</i> . <i>Journal of Double Star Observations</i> , 11(1s).	
Spectroscopy Research <i>University of South Alabama</i>	2011 – 2012

Used gamma-ray spectroscopy to date soil core samples via Cs-137, Pb-210, and other radioactive isotopes, under Dr. Justin Sanders.